

PYTHON INTERVIEW QUESTIONS

Crack Python Interviews With Confidence

20 real interview questions - verified, solved, and explained



**From real interviews at Infosys, Wipro, Cognizant,
Deloitte, Microsoft and more**

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EASY (Fundamentals)

1) What is the difference between a list and a tuple in Python?

Infosys

- A:
- **List** = Mutable, Slower, Uses [], Data can change.
 - **Tuple** = Immutable, Faster, Uses (), Data stays constant.

2) What are the key features of Python?

HARMAN

A: Python is:

- Interpreted and dynamically typed
- Easy to read, write, and learn
- Open-source and cross-platform
- Supports multiple programming paradigms
- Rich in libraries and frameworks

3) What is the difference between the '==' operator and the 'is' operator in Python?

Infosys

- A:
- We use "==" to check whether two objects have the same value (content equality).
 - We use "is" to check whether two references point to the same object in memory (identity).

```
list = [10, 20]
```

```
list == list[:]          #True - same values
```

```
list is list[:]          #False - different objects in memory
```

4) Could you explain list comprehension and write examples?

Infosys

- A:
- List comprehension lets you create a new list in a short, clean, single-line expression.

```
evens = [x for x in range(10) if x % 2 == 0]
print(evens) # [0, 2, 4, 6, 8]
```

5) What are mutable vs. immutable data types in Python? (e.g. why a tuple is immutable, a list isn't)

Cognizant

- A:
- Mutable types can be changed after creation (e.g., list, dict, set).
 - Immutable types cannot be changed; any update creates a new object (e.g., tuple, string, int).

Why is "tuple" immutable? - Because its structure is fixed.

Why is "list" mutable? - Because it is designed to allow adding or modifying items.

6) How to reverse a number, string conversions in Python?

Cognizant

- A:
- To reverse a number, convert it to a string, reverse the string, and convert it back to a number.
 - To reverse a string, use slicing (`[::-1]`) to reverse the characters.

7) What are Python decorators?

Infosys

A: Python decorators are functions that modify or extend the behaviour of other functions without changing their actual code.

8) How do you sort a list in Python?

Infosys

A: Sorting a list means arranging its elements in order.

- In Python, you can use `sort()` to sort the list itself or `sorted()` to get a new sorted list.

```
nums = [5, 2, 8, 1]
nums.sort()           # [1, 2, 5, 8]

sorted_nums = sorted(nums, reverse=True) # [8, 5, 2, 1]
```

9) How do you separate text and numbers from a mixed string?

Infosys

A: • To separate letters and numbers from a string, iterate through each character and use `isalpha()` for letters and `isdigit()` for numbers.

```
s = "abc123"
text = "".join([c for c in s if c.isalpha()])
numbers = "".join([c for c in s if c.isdigit()])

print(text)      # abc
print(numbers)   # 123
```

10) What is PEP 8, and why is it important?

Barclays

A: PEP 8 is the official Python Style Guide that defines best practices for writing clean, readable, and consistent code. Example rule: Use 4 spaces per indentation level.

11) What is the difference between break, continue, and pass?

Wipro

- A:
- break → Exits the loop completely.
 - continue → Skips the current iteration and moves to the next one.
 - pass → Does nothing; used as a placeholder in code.

```
for i in range(5):  
    if i == 2:  
        continue    # skip 2  
    if i == 4:  
        break        # stop loop  
    pass             # placeholder  
    print(i)         # Output: 0, 1, 3
```

12) What is a lambda function?

Deloitte

- A:
- A lambda function is an anonymous, one-line function used for simple, short operations. It doesn't need a name and is often used for short, throwaway functions.

```
add = lambda x, y: x + y  
print(add(3, 4))    # 7
```

MEDIUM (Intermediate)

1) Explain list slicing with examples.

Tech Mahindra

A: List slicing is used to access a subset of elements from a list using the [start:stop:step] syntax.

```
nums = [0, 1, 2, 3, 4, 5]
```

```
print(nums[1:4])    # [1, 2, 3]    → from index 1 to 3
print(nums[:3])     # [0, 1, 2]    → from start to index 2
print(nums[::2])    # [0, 2, 4]    → every 2nd element
print(nums[::-1])   # [5, 4, 3, 2, 1, 0] → reversed list
```

2) What is the difference between deepcopy and shallow copy?

SAP

A: Shallow Copy: Creates a new object but only copies references to nested objects. Changes in nested objects affect the original.

- Example: `b = a.copy()`

Deep Copy: Creates a completely independent copy, including all nested objects. Changes do not affect the original.

```
import copy
a = [[1, 2], [3, 4]]
b = copy.deepcopy(a)    # deep copy
```


3) How do you print all permutations of a string in Python?

Microsoft

A: You can generate permutations using recursion, where you fix one character at a time and permute the rest.

```
def permute(s, path=""):
    if not s:
        print(path)
        return
    for i in range(len(s)):
        permute(s[:i] + s[i+1:], path + s[i])

permute("abc")
```

4) How do you count the number of vowels in a string in Python?

Deloitte

A:

```
def count_vowels(s):
    return sum(1 for ch in s.lower() if ch in "aeiou")

print(count_vowels("Hello World")) # Output: 3
```

5) How do you check if a string is a palindrome in Python?

Deloitte

A: A palindrome is a string that reads the same forward and backward.

```
def is_palindrome(s):
    s = s.lower().replace(" ", "") # normalize
    return s == s[::-1]

print(is_palindrome("Madam")) # True
```

Naveen Dhawan

6) Write a Python program to print all prime numbers between 1 and 100.

Cognizant

A:

```
for num in range(2, 101):
    is_prime = True
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
    if is_prime:
        print(num)
```

7) Write a Python program to count how many times each character appears in the string "Data Analysis".

Wipro

A:

```
def char_frequency(s):
    freq = {}
    for ch in s:
        freq[ch] = freq[ch] + 1 if ch in freq else 1
    return freq

print(char_frequency("Data Analysis"))
```

8) Given a nested list [[1,2], [3,4], [[4], [6]]], write a Python program to flatten it into a single list: [1,2,3,4,4,6].

Wipro

Your turn to solve this one — try it out!